

1. Which description best describes a correlation coefficient of $r = -0.62$?
 - (a) Weakly negative relationship
 - (b) Strong negative relationship
 - (c) No relationship exists between the variables
 - (d) A statistically insignificant relationship
2. Using the U.S. General Social Survey data, a researcher suspects Americans are watching more tv today than in the 1970s. Assuming an α level of .05, should the researcher reject the null hypothesis? Why or why not?

Welch Two Sample t-test

data: my_data_{tvhours} by my_data_{year} $t = -3$, $df = 3617$, $p\text{-value} = 0.005$ alternative hypothesis: true difference in means between group 1975 and group 2024 is not equal to 0 95 percent confidence interval: -0.432 -0.079 sample estimates: mean in group 1975 mean in group 2024 3.0 3.3

- (a) No, because the p-value (0.005) is greater than the α level of .05.
 - (b) No, because the p-value (0.005) equals the α level of .05.
 - (c) Yes, because the p-value (0.005) is greater than the α level of .05.
 - (d) Yes, because the p-value (0.005) is less than the α level of .05.
3. If a researcher concluded that there was no difference in police use of force against Black and White men when there actually is a difference, then:
 - (a) The null hypothesis was correctly rejected.
 - (b) The null hypothesis was incorrectly rejected (false positive).
 - (c) The null hypothesis was correctly retained.
 - (d) The null hypothesis was incorrectly retained (false negative).
4. Using the U.S. General Social Survey data, a researcher tests whether the proportions who favor a law requiring a person to obtain a police permit before they could buy a gun have changed over time. Assuming an α level of .05, which of these statements best summarizes the researcher's conclusion?

Welch Two Sample t-test

data: my_data_{gunlaw} by my_data_{year} $t = 2$, $df = 3115$, $p\text{-value} = 0.07$ alternative hypothesis: true difference in means between group 1984 and group 2024 is not equal to 0 95 percent confidence interval: -0.0022 0.0581 sample estimates: mean in group 1984 mean in group 2024 0.72 0.70

- (a) There is not a statistically significant difference in support for the police permit law between 1984 and 2024, indicating that public opinion has remained stable.
 - (b) The difference in support for the police permit law between 1984 and 2024 was due to sampling error and cannot be interpreted.
 - (c) There is a statistically significant increase in support for the police permit law from 1984 to 2024, proving that stricter gun control laws are now favored by most Americans.
 - (d) There is a statistically significant difference in support for the police permit law between 1984 and 2024, suggesting that public opinion has changed over time.
5. What key concept does Wheelan illustrate in the chapter titled Inference: Why My Statistics Professor Thought I Might Have Cheated?
 - (a) That statistical inference can be used to prove someone's guilt beyond reasonable doubt
 - (b) That hypothesis testing is not useful in criminal investigations
 - (c) That cheating can be detected by comparing mean scores across different groups
 - (d) That unusually accurate results can raise suspicion, but statistical inference helps assess whether they are likely due to chance

6. Which of these datasets has a standard deviation of 0?

- (a) 0, 2, 4, 6, 8
- (b) 1, 4, 7, 10, 15
- (c) 4, 4, 4, 4, 4
- (d) 2, 2, 5, 7, 7

7. A researcher used the 2024 General Social Survey to find the proportion of the U.S. population that reports believing in some form of life after death. Based on the R output with 90% confidence intervals, which of the following is a correct interpretation of their results?

n	mean	sd	lowCI	hiCI
1283	0.83	0.38	0.81	0.85

- (a) About 90% of people responded to the national survey.
 - (b) Because the mean is within the confidence interval, there is a .90 probability that the proportion of U.S. adults who believe in some form of life after death is .83.
 - (c) The researcher is 90% confident that the proportion of U.S. adults who believe in some form of life after death is between .81 and .85.
 - (d) If another survey had been conducted in 2024, there's a 90% chance the point estimate would be between .81 and .85.
8. Using the table, which of the following is the correct formula to find the expected frequency for U.S. liberals who oppose the death penalty in 2024?

ideology	conservative	liberal	Total
cappun favor	512 (66.8%)	255 (33.2%)	767 (100.0%)
oppose	168 (32.7%)	345 (67.3%)	513 (100.0%)
Total	680 (53.1%)	600 (46.9%)	1280 (100.0%)

- (a) $(600 * 513)/1280$
 - (b) $(345 * 600)/1280$
 - (c) $(600 * 767)/1280$
 - (d) $(345 * 513)/1280$
9. According to the 68-95-99.7 rule, what can we say about a person whose score is more than 2 standard deviations above the mean?
- (a) Their score is unusual and higher than most.
 - (b) Their score is likely below average.
 - (c) Their score is within the middle 68% of the distribution.
 - (d) Their score is very common.
10. Researchers found that couples report about 10 arguments per month on average. If a couple received a z-score of -2 for this measure, which of the following best describes the amount of arguments for this couple compared with the average?
- (a) This couple had fewer arguments than about 2% of couples.
 - (b) This couple had fewer arguments than about 98% of couples.
 - (c) This couple had fewer arguments than about 5% of couples.
 - (d) This couple had about 8 arguments per month.

11. In a normally distributed set of test scores, a student scoring below the average will have a _____.
 (a) positive standard error
 (b) negative standard error
 (c) positive z-score
 (d) negative z-score
12. What does the expression `5 != 3` evaluate to in R?
 (a) TRUE
 (b) FALSE
 (c) NA
 (d) NULL
13. In order to compare meaningfully across groups, a frequency should be converted into a/an _____.
 (a) number
 (b) percentage
 (c) ordinal variable
 (d) standard deviation
14. What does the following code return?
- ```
gss24 |>
 group_by(sex) |>
 summarize(mean_age = mean(age, na.rm = TRUE))
```
- (a) A filtered data frame with only rows where age is not missing.  
 (b) A new column named `mean_age` added to the original data frame.  
 (c) a data frame showing the average age for each sex.  
 (d) a list of all ages grouped by sex.
15. What does Oscar Gandy suggest about the use of racial statistics?  
 (a) They are neutral tools most useful for academic research  
 (b) They can both expose and perpetuate discrimination  
 (c) Their use in legal proceedings increasingly lead to unfair policy outcomes  
 (d) Their use in data journalism helps mitigate online polarization
16. Which of the following statements best reflects the ASA's guidance on interpreting p-values?  
 (a) P-values measure the scientific or economic significance of a result.  
 (b) A large p-value proves that no effect exists.  
 (c) P-values are influenced by sample size and measurement precision, and do not reflect effect size or importance.  
 (d) A small p-value always indicates a large and important effect.
17. Why is it considered good practice to use RStudio Projects when working on data analysis?  
 (a) They ensure your code runs faster by optimizing memory usage  
 (b) They automatically load packages when you open RStudio, so there's no need to use `library()`  
 (c) They help organize your files, set a working directory, and keep your environment consistent  
 (d) They prevent syntax errors by enforcing tidyverse formatting rules
18. The level of measurement for the `hapmar` variable is \_\_\_\_\_.  
 [1] "Happiness of marriage" Frequencies  
`gss24$hapmar`  
 Type: Factor

|               | Freq | %      | % Cum. |
|---------------|------|--------|--------|
| very happy    | 802  | 59.36  | 59.36  |
| pretty happy  | 504  | 37.31  | 96.67  |
| not too happy | 45   | 3.33   | 100.00 |
| Total         | 1351 | 100.00 | 100.00 |

- (a) interval
- (b) dichotomous
- (c) nominal
- (d) ordinal

19. Which command installs a new package in R?

- (a) `require()`
- (b) `load()`
- (c) `install.packages()`
- (d) `library()`

20. A researcher used the 2024 General Social Survey to find the proportion of the U.S. population that reports believing in some form of life after death. Based on the R output of their findings, with 95% confidence intervals, which of the following is a correct interpretation of their results?

| sex    | n    | mean | sd  | lowCI | hiCI |
|--------|------|------|-----|-------|------|
| male   | 1139 | 4.2  | 3.7 | 4.0   | 4.5  |
| female | 1174 | 4.0  | 3.5 | 3.8   | 4.2  |

- (a) The confidence intervals do not overlap so we can conclude that the average time men and women spend on leisure activities per day are statistically significantly different.
- (b) We cannot determine whether the null hypothesis of zero difference between the parameters will be rejected at the 0.05 level.
- (c) We are 95% certain that women have less time per day than men for leisure activities, likely because of their greater time spent on housework and childcare.
- (d) The confidence intervals overlap, so we can't know for sure if there is a statistical difference between the average time men and women spend on leisure activities per day.

21. If a statistics classes' final grades were normally distributed with an average of 81.5% and a standard deviation of 1.5, about what proportion of the students' final grades were between 80% and 83%?

- (a) 50%
- (b) 68%
- (c) 62%
- (d) 95%

22. National survey data shows that 69% of people support mandatory childhood vaccinations. A local researcher believes that support is likely much higher in Toronto compared with the national average. Which of the following equations represents the research hypothesis?

- (a)  $\mu \neq 0.69$
- (b)  $\mu < 0.69$
- (c)  $\mu = 0.69$
- (d)  $\mu > 0.69$

23. Which RStudio pane is used to write and edit R scripts and Quarto documents?
- (a) Console
  - (b) Source
  - (c) Files
  - (d) Environment
24. In *Naked Statistics*, Charles Wheelan explains that “most sample means will lie reasonably close to the population mean; the standard error is what defines ‘reasonably close.’” What does this statement imply?
- (a) The standard error measures how much sample means tend to vary around the population mean.
  - (b) The sample mean is always equal to the population mean.
  - (c) The standard error increases as sample size increases.
  - (d) The standard error tells us how far the population mean is from the sample mean.
25. What will happen if you try to use a function from a package that hasn’t been loaded?
- (a) You will get an error saying the function is not found
  - (b) R will prompt you to install the package
  - (c) R will automatically load the package
  - (d) The function will run normally
26. Researchers calculated a 95% confidence interval of the average family size of Canadians to be from 2.7 to 3.1, based on a sample of 500 families with a standard deviation of 1.1. What change would produce a more precise estimate?
- (a) Improving our sampling strategy
  - (b) Increasing our sample size to 600
  - (c) Increasing our standard deviation to 1.3
  - (d) Increasing our confidence interval to 99%
27. Chi-square values cannot be \_\_\_\_.
- (a) positive
  - (b) zero
  - (c) outside the range of 0 to 1
  - (d) negative
28. What is the result of the following code?
- ```
gss24 |>
  filter(age > 20) |>
  select(age, childs)
```
- (a) rows with `age > 20` are removed
 - (b) The original `gss24` dataset is overwritten.
 - (c) Only rows with `age > 20` and columns `age` and `childs` are returned.
 - (d) A new column is added to `gss24`
29. What is Wheelan’s primary message of Chapter 1 (“What’s the Point?”) in *Naked Statistics: Stripping the Dread from the Data*?
- (a) Statistics explain why relationship exist between variables
 - (b) Statistics provide honest answers to complex questions
 - (c) Statistical formulas are the most important part of understanding data
 - (d) Statistics simplify complex information to help us make better decisions
30. Based on the table below, which statement best compares the relative frequency of being `not too happy` between men and women?

Cross-Tabulation, Column Proportions
happy * sex
Data Frame: gss24

	sex	male	female	Total
happy				
very happy		293 (20.2%)	386 (21.3%)	679 (20.8%)
pretty happy		830 (57.1%)	1054 (58.2%)	1884 (57.7%)
not too happy		330 (22.7%)	370 (20.4%)	700 (21.5%)
Total		1453 (100.0%)	1810 (100.0%)	3263 (100.0%)

- (a) The frequency of being `not too happy` is equal for men and women.
 - (b) Men have a slightly higher relative frequency of being `not too happy` than women, even though fewer men reported it.
 - (c) A greater number of women reported being `not too happy` than men, so women have a higher relative frequency.
 - (d) Women have a slightly higher relative frequency because more women than men reported being `not too happy`.
31. What is the purpose of `drop_na()` in `dplyr`?
- (a) It removes rows that contain missing values.
 - (b) It converts NA values to NULL.
 - (c) It drops columns that are entirely NA.
 - (d) It replaces missing values with zeros.
32. Which of the following best describes the differences between the Console, R Scripts, and Quarto documents in RStudio?
- (a) The Console is used for writing long-term code, R Scripts are for temporary testing, and Quarto is only for plotting.
 - (b) R Scripts and Quarto documents both run code line-by-line, but only the Console saves your work.
 - (c) The Console and R Scripts are identical, while Quarto is used only for Markdown formatting.
 - (d) The Console runs code interactively, R Scripts store reusable code, and Quarto combines code, text, and output for dynamic reports.
33. According to Cohen's argument, what role should descriptive statistics play in social science research?
- (a) Descriptive statistics are outdated and should be replaced by more advanced inferential methods.
 - (b) Descriptive statistics are useful only as a predominately as a step before hypothesis testing and theory development.
 - (c) Descriptive work should be minimized in favor of studies that make strong theoretical contributions.
 - (d) Descriptive statistics are a foundational part of scientific inquiry and should be valued as legitimate research.
34. According to Wheelan, what is the most common cause of inaccurate polling results?
- (a) Misuse of statistical software during data analysis
 - (b) Incorrect calculations of standard errors
 - (c) Random variation in respondent answers
 - (d) Biased sampling methods or poorly worded questions
35. The variable `mntlhlth` measures how many days in the past 30 a respondent reported poor mental health. A respondent reports 5 poor mental health days. Based on the summary statistics for this variable, which of the following best describes their z-score?
- (a) The z-score is large and positive because 5 is much higher than the mean.
 - (b) The z-score is undefined because the median is 0.
 - (c) The z-score is negative because 5 is below the sd.
 - (d) The z-score is close to zero because 5 is close to the mean.

freq	median	mean	sd
2279	0	4.5	7.9

36. Has public opinion changed about whether Black Americans have worse jobs, income, and housing than white Americans mainly because of discrimination? Based on a significance level of $\alpha = 0.01$, what's the best conclusion from the test?

	year		Total
	2022	2024	
opinion			
no	509 (45%)	522 (51%)	1,031 (48%)
yes	619 (55%)	494 (49%)	1,113 (52%)
Total	1,128 (100%)	1,016 (100%)	2,144 (100%)

Pearson's Chi-squared test, p=0.004

- (a) There is no statistically significant difference in attitudes between 2022 and 2024.
 (b) The results show that most people in both years agreed racial inequality is due to discrimination.
 (c) The p-value of .000 means the null hypothesis is true.
 (d) The evidence shows attitudes about racial inequality differ between 2022 and 2024.
37. What does the pipe operator (`|>`) do in dplyr?
- (a) It passes the result of one function into the next.
 (b) It creates a new column in a data frame.
 (c) It loads the dplyr package into memory.
 (d) It filters rows based on a condition.
38. What does the following R code produce?
- ```
my_data |>
 tbl_cross(
 row = incgap,
 col = sex,
 percent = "column",
 missing = "no") |>
 add_p(source_note = TRUE)
```
- (a) A cross-tabulation of `incgap` by `sex`, showing column percentages and a chi-square test of independence  
 (b) A frequency table of `incgap` responses with missing values imputed and grouped by `sex`  
 (c) A regression output predicting `incgap` from `sex`, including R-squared and p-values  
 (d) A summary table comparing the mean values of `incgap` between men and women using a t-test
39. A p-value is
- (a) A measure that tells us the variability in a normal distribution  
 (b) An estimate of the standard error of a sampling distribution  
 (c) An estimate of where our population statistic lies on the true sampling distribution of the means  
 (d) A measure of how unusual or rare our obtained statistic is compared with what is stated in the null hypothesis
40. The U.S. General Social Survey asked people how many brothers and sisters they had. A researcher used R to produce the summary statistics of this variable (`sibs`) below. Based on this, you conclude the distribution for this variable is:

| freq | median | mean | sd  |
|------|--------|------|-----|
| 3291 | 3      | 2.9  | 1.8 |

- (a) nominal
  - (b) symmetrical
  - (c) negatively skewed
  - (d) positively skewed
41. Which description best describes the correlation between personal income (`realrinc`) and number of days of poor mental health (`mntlhlth`)?
- Pearson's product-moment correlation
- data: `realrinc` and `mntlhlth`  $t = -5$ ,  $df = 1933$ ,  $p\text{-value} = 5e-07$  alternative hypothesis: true correlation is not equal to 0 95 percent confidence interval: -0.16 -0.07 sample estimates: `cor` -0.11
- (a) There is no evidence of an association between personal income and days of poor mental health ( $t = -5$ ).
  - (b) Results indicated an inverse relationship between personal income and days of poor mental health ( $r = -0.11$ ,  $p < .05$ ), suggesting that the number of days of poor mental health decreases as personal income increases.
  - (c) Decreases in personal income was correlated with decreases in the number of days of poor mental health ( $t = -5$ ,  $p < .05$ ).
  - (d) There is a weakly positive association between personal income and days of poor mental health ( $r = -0.11$ ,  $p < .05$ ).
42. What does the `::` operator do in R?
- (a) Specifies a function from a particular package (without loading it)
  - (b) Assigns a value to a variable
  - (c) Comments out a line of code
  - (d) Imports all functions from a package
43. When examining a frequency distribution, which of the following columns generally allows you to identify the median most quickly?
- (a) cumulative percentages
  - (b) frequencies
  - (c) percentages
  - (d) cumulative frequencies
44. Which of the following is a measure of the spread of a distribution?
- (a) median
  - (b) mean
  - (c) standardized variable
  - (d) standard deviation
45. Which function is used to load a package in R?
- (a) `source()`
  - (b) `load()`
  - (c) `install.packages()`
  - (d) `library()`
46. Which of the following code snippets correctly uses `mutate()` to convert height from centimeters to meters?
- (a) `mutate(height = select(height / 100))`
  - (b) `mutate(height / 100 = height)`
  - (c) `mutate(height = height / 100)`
  - (d) `mutate(height <- height / 100)`



47. Which of the following is a valid use of `group_by()` and `summarize()`?

- (a) `summarize(group_by(data, age), mean(sex))`
- (b) `data |> summarize(mean = group_by(sex))`
- (c) `group_by(data, summarize(sex))`
- (d) `data |> group_by(sex) |> summarize(freq = n())`

48. A sociologist wants to examine whether there is a relationship between level of education (measured in years) and annual income (in dollars) among adults in the U.S. Which statistical test is most appropriate?

- (a) Logistic regression
- (b) Chi-square test
- (c) Pearson's correlation (r)
- (d) t-test

49. What will the following code do?

```
gss24 <- gss24 |>
 mutate(status = case_when(
 income > 100000 ~ "High",
 income > 50000 ~ "Medium",
 TRUE ~ "Low"
))
```

- (a) It replaces the numeric values in the income column with text labels.
- (b) It creates a new column `status` with 3 possible values.
- (c) It filters out rows where income is less than 50,000.
- (d) It removes missing values from the income column.

50. A researcher compares support for universal healthcare (Support vs. Oppose) between two racial groups: Black and White respondents. Which statistical test should be used?

- (a) Linear regression
- (b) t-test
- (c) Pearson's correlation (r)
- (d) Chi-square test

1. (a) ☐ (b) ☒ (c) ☐ (d) ☐
2. (a) ☐ (b) ☐ (c) ☐ (d) ☒
3. (a) ☐ (b) ☐ (c) ☐ (d) ☒
4. (a) ☒ (b) ☐ (c) ☐ (d) ☐
5. (a) ☐ (b) ☐ (c) ☐ (d) ☒
6. (a) ☐ (b) ☐ (c) ☒ (d) ☐
7. (a) ☐ (b) ☐ (c) ☒ (d) ☐
8. (a) ☒ (b) ☐ (c) ☐ (d) ☐
9. (a) ☒ (b) ☐ (c) ☐ (d) ☐
10. (a) ☐ (b) ☒ (c) ☐ (d) ☐
11. (a) ☐ (b) ☐ (c) ☐ (d) ☒
12. (a) ☒ (b) ☐ (c) ☐ (d) ☐
13. (a) ☐ (b) ☒ (c) ☐ (d) ☐
14. (a) ☐ (b) ☐ (c) ☒ (d) ☐
15. (a) ☐ (b) ☒ (c) ☐ (d) ☐
16. (a) ☐ (b) ☐ (c) ☒ (d) ☐
17. (a) ☐ (b) ☐ (c) ☒ (d) ☐
18. (a) ☐ (b) ☐ (c) ☐ (d) ☒
19. (a) ☐ (b) ☐ (c) ☒ (d) ☐
20. (a) ☐ (b) ☐ (c) ☐ (d) ☒
21. (a) ☐ (b) ☒ (c) ☐ (d) ☐
22. (a) ☐ (b) ☐ (c) ☐ (d) ☒

23. (a) ☐ (b) ☒ (c) ☐ (d) ☐
24. (a) ☒ (b) ☐ (c) ☐ (d) ☐
25. (a) ☒ (b) ☐ (c) ☐ (d) ☐
26. (a) ☐ (b) ☒ (c) ☐ (d) ☐
27. (a) ☐ (b) ☐ (c) ☐ (d) ☒
28. (a) ☐ (b) ☐ (c) ☒ (d) ☐
29. (a) ☐ (b) ☐ (c) ☐ (d) ☒
30. (a) ☐ (b) ☒ (c) ☐ (d) ☐
31. (a) ☒ (b) ☐ (c) ☐ (d) ☐
32. (a) ☐ (b) ☐ (c) ☐ (d) ☒
33. (a) ☐ (b) ☐ (c) ☐ (d) ☒
34. (a) ☐ (b) ☐ (c) ☐ (d) ☒
35. (a) ☐ (b) ☐ (c) ☐ (d) ☒
36. (a) ☐ (b) ☐ (c) ☐ (d) ☒
37. (a) ☒ (b) ☐ (c) ☐ (d) ☐
38. (a) ☒ (b) ☐ (c) ☐ (d) ☐
39. (a) ☐ (b) ☐ (c) ☐ (d) ☒
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43. (a) ☒ (b) ☐ (c) ☐ (d) ☐
44. (a) ☐ (b) ☐ (c) ☐ (d) ☒

45. (a) ☐ (b) ☐ (c) ☐ (d) ☒
46. (a) ☐ (b) ☐ (c) ☒ (d) ☐
47. (a) ☐ (b) ☐ (c) ☐ (d) ☒
48. (a) ☐ (b) ☐ (c) ☒ (d) ☐
49. (a) ☐ (b) ☒ (c) ☐ (d) ☐
50. (a) ☐ (b) ☐ (c) ☐ (d) ☒